

Time-Resolved Asymmetry: Human-AI vs Human-Human vs AI-AI

How story-internal asymmetry evolves across early / middle / late phases

PENPAL Project

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1 Overview

This notebook tracks how within-dyad asymmetry evolves *across the story*. For each metric (valence, novelty-resonance slope, semantic-exploration AUC), we compute the per-story signed and unsigned slot-1 vs slot-2 difference within rolling thirds (Early / Middle / Late) of the story turn window, and test whether condition $\times$ phase interactions appear.

Two outcomes per metric:

- **Magnitude:** per-story $|A|$ in each phase.
- **Direction:** per-story signed slot-difference in each phase (with the identity-tied directionality reading: HA should keep a stable sign across phases; HH and AA should show no fixed sign).

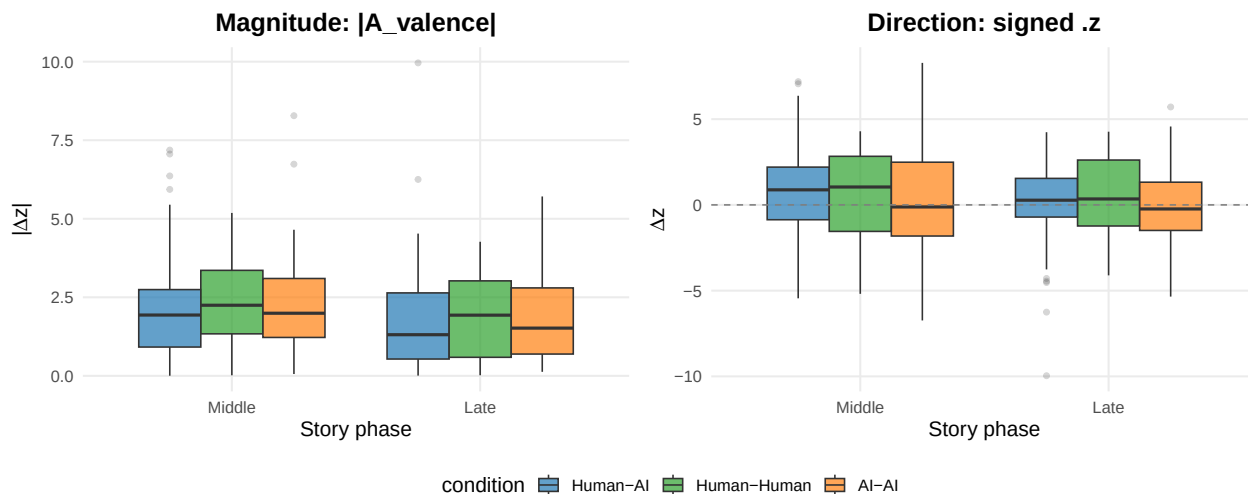
2 Phase definition

3 Valence Asymmetry by Phase

Table 1: Per-story valence asymmetry by condition and story phase

condition	phase	n	mean_signed	sd_signed	mean_abs
Human-AI	Middle	98	0.746	2.507	2.064
Human-AI	Late	99	0.243	2.354	1.730
Human-Human	Middle	36	0.741	2.583	2.324
Human-Human	Late	36	0.418	2.272	1.873
AI-AI	Middle	40	0.427	2.918	2.378
AI-AI	Late	40	-0.113	2.393	1.860

condition	phase	n	mean_signed	sd_signed	mean_abs
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```
## LMM: A_valence ~ condition * phase + (1 | conversation_id)
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: A_valence ~ condition * phase + (1 | conversation_id)
## Data: valence_phase_A
##
## REML criterion at convergence: 1294.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.4984 -0.7444 -0.1516  0.5833  5.2516
##
## Random effects:
##  Groups             Name             Variance Std.Dev.
##  conversation_id (Intercept) 0.05421  0.2328
##  Residual                2.32750  1.5256
## Number of obs: 349, groups: conversation_id, 176
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      2.0638    0.1559 342.8399  13.239  <2e-16
## conditionHuman-Human      0.2600    0.3008 342.8299   0.864   0.388
## conditionAI-AI           0.3139    0.2896 342.8302   1.084   0.279
## phaseLate           -0.3336    0.2174 174.4359  -1.534   0.127
## conditionHuman-Human:phaseLate -0.1170    0.4202 172.5059  -0.278   0.781
## conditionAI-AI:phaseLate  -0.1836    0.4045 172.5614  -0.454   0.650
##
## (Intercept)                ***
## conditionHuman-Human
## conditionAI-AI
## phaseLate
## conditionHuman-Human:phaseLate
## conditionAI-AI:phaseLate
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```

##
## Correlation of Fixed Effects:
##      (Intr) cndH-H cnAI-AI phasLt cH-H:L
## cndtnHmn-Hm -0.518
## condtnAI-AI -0.538  0.279
## phaseLate   -0.701  0.363  0.377
## cndtnHm-H:L  0.363 -0.700 -0.195  -0.517
## cndtAI-AI:L  0.377 -0.195 -0.700  -0.537  0.278
##
## LMM: signed_diff ~ condition * phase + (1 | conversation_id)
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: signed_diff ~ condition * phase + (1 | conversation_id)
##   Data: valence_phase_A
##
## REML criterion at convergence: 1622.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.1027 -0.6577  0.0028  0.6937  3.1578
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 0.000      0.000
## Residual                  6.186      2.487
## Number of obs: 349, groups: conversation_id, 176
##
## Fixed effects:
##              Estimate Std. Error      df t value
## (Intercept)    0.745545   0.251250 343.000000    2.967
## conditionHuman-Human -0.004526   0.484738 343.000000   -0.009
## conditionAI-AI      -0.318755   0.466676 343.000000   -0.683
## phaseLate        -0.502982   0.354423 343.000000   -1.419
## conditionHuman-Human:phaseLate 0.179685   0.685058 343.000000    0.262
## conditionAI-AI:phaseLate -0.037006   0.659497 343.000000   -0.056
##              Pr(>|t|)
## (Intercept)    0.00321 **
## conditionHuman-Human 0.99256
## conditionAI-AI      0.49505
## phaseLate        0.15676
## conditionHuman-Human:phaseLate 0.79325
## conditionAI-AI:phaseLate 0.95528
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) cndH-H cnAI-AI phasLt cH-H:L
## cndtnHmn-Hm -0.518
## condtnAI-AI -0.538  0.279
## phaseLate   -0.709  0.367  0.382
## cndtnHm-H:L  0.367 -0.708 -0.197  -0.517
## cndtAI-AI:L  0.381 -0.197 -0.708  -0.537  0.278
## optimizer (nloptwrap) convergence code: 0 (OK)

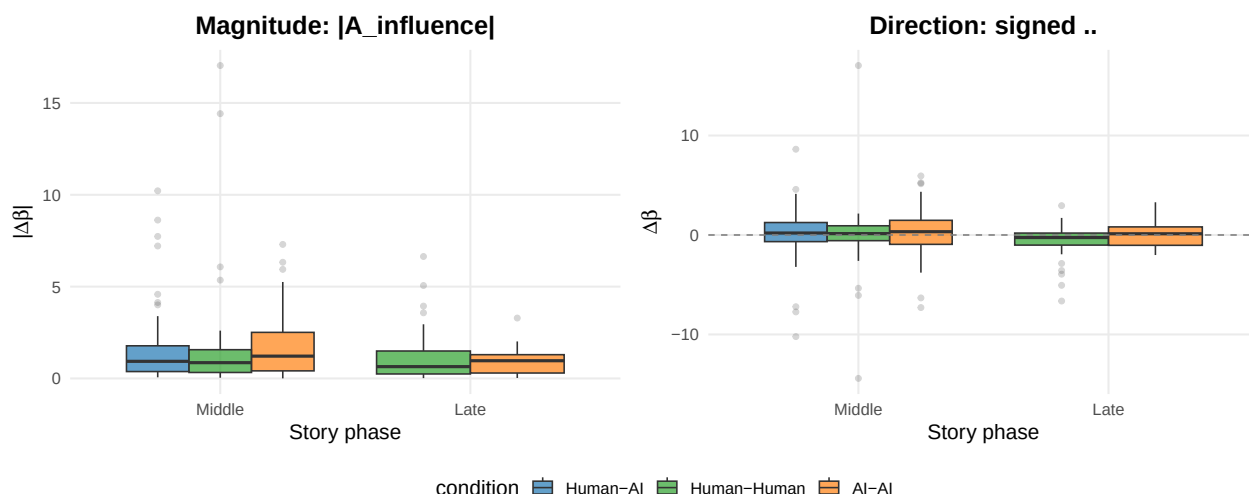
```

```
## boundary (singular) fit: see help('isSingular')
```

4 Novelty-Resonance Slope Asymmetry by Phase

Table 2: Per-story novelty-resonance slope asymmetry by condition and story phase

condition	phase	n	mean_signed	sd_signed	mean_abs
Human-AI	Middle	98	0.184	2.303	1.468
Human-Human	Middle	36	-0.109	4.150	1.968
Human-Human	Late	36	-0.630	1.875	1.245
AI-AI	Middle	40	0.184	2.727	1.910
AI-AI	Late	40	0.012	1.171	0.927



```
## LMM: A_influence ~ condition * phase + (1 | conversation_id)
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: A_influence ~ condition * phase + (1 | conversation_id)
## Data: novelty_phase_A
##
## REML criterion at convergence: 1060.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -0.9950 -0.5130 -0.2295  0.1926  6.9687
##
## Random effects:
## Groups           Name          Variance Std.Dev.
## conversation_id (Intercept) 0.6305  0.794
## Residual                3.5091  1.873
## Number of obs: 250, groups: conversation_id, 174
##
## Fixed effects:
##              Estimate Std. Error    df t value Pr(>|t|)
## (Intercept)      1.4680     0.2055 242.4111   7.143 1.07e-11
```

```

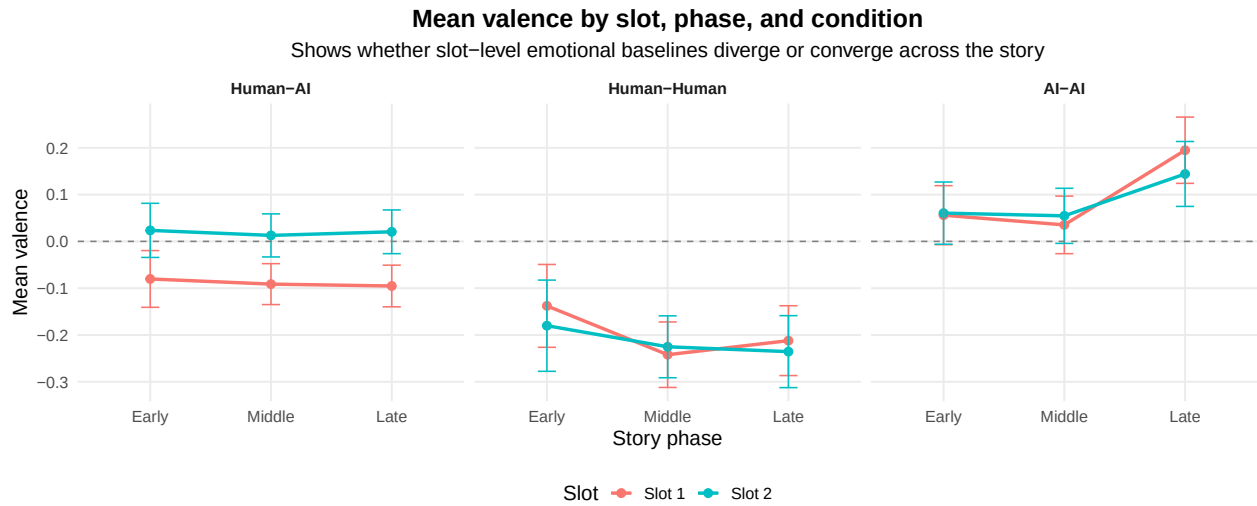
## conditionHuman-Human          0.5004      0.3965 242.4111    1.262    0.208
## conditionAI-AI                0.4416      0.3817 242.4111    1.157    0.248
## phaseLate                    -0.9824      0.4189 101.3123   -2.345    0.021
## conditionHuman-Human:phaseLate 0.2592      0.6086 101.3123    0.426    0.671
##
## (Intercept)                    ***
## conditionHuman-Human
## conditionAI-AI
## phaseLate                      *
## conditionHuman-Human:phaseLate
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) cndH-H cAI-AI phasLt
## cndtnHmn-Hm -0.518
## condtnAI-AI -0.538  0.279
## phaseLate   0.000  0.000 -0.549
## cndtnHm-H:L 0.000 -0.404  0.378 -0.688
## fit warnings:
## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
##
## LMM: signed_diff ~ condition * phase + (1 | conversation_id)
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: signed_diff ~ condition * phase + (1 | conversation_id)
## Data: novelty_phase_A
##
## REML criterion at convergence: 1170.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -5.1782 -0.2643  0.0440  0.3938  6.4996
##
## Random effects:
## Groups          Name          Variance Std.Dev.
## conversation_id (Intercept) 0.6292   0.7932
## Residual                5.8308   2.4147
## Number of obs: 250, groups: conversation_id, 174
##
## Fixed effects:
##              Estimate Std. Error      df t value
## (Intercept)    0.184336   0.256745 243.869637    0.718
## conditionHuman-Human -0.293324   0.495339 243.869637   -0.592
## conditionAI-AI      -0.000656   0.476882 243.869637   -0.001
## phaseLate       -0.171804   0.539942 100.237183   -0.318
## conditionHuman-Human:phaseLate -0.348889   0.784518 100.237183   -0.445
##
##              Pr(>|t|)
## (Intercept)    0.473
## conditionHuman-Human 0.554
## conditionAI-AI    0.999
## phaseLate       0.751
## conditionHuman-Human:phaseLate 0.657

```

```
##
## Correlation of Fixed Effects:
##          (Intr) cndH-H cAI-AI phasLt
## cndtnHmn-Hm -0.518
## condtnAI-AI -0.538  0.279
## phaseLate    0.000  0.000 -0.566
## cndtnHm-H:L  0.000 -0.417  0.390 -0.688
## fit warnings:
## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
```

5 Mean Valence Trajectory by Slot and Condition

A descriptive companion view: how does each slot’s mean valence drift across phase, by condition? Useful for the “story-evolution” narrative without relying on the asymmetry abstraction.



6 Summary

What this notebook adds beyond the cross-section comparisons:

- **Per-phase magnitude $|A|$** for valence and novelty-resonance: tests whether asymmetry strengthens, weakens, or stays flat across the story.
- **Per-phase signed asymmetry:** combined with per-story directional-stability, this is the strongest visual evidence for HA-specific identity-tied directionality. A signed-asymmetry boxplot that stays *consistently above zero* in HA across all three phases, while HH/AA hover around zero, is the publication-quality figure for the reframed RQ1.
- **Mean valence trajectory by slot and condition:** a story-evolution view that does not depend on the asymmetry abstraction and helps illustrate qualitative drift.

Open additions worth considering:

- Add a per-phase semantic-exploration AUC view (the lag k axis is itself an evolution axis, but a phase split could surface fine-grained shifts).
- Replace fixed thirds with overlapping rolling windows for smoother trajectories.

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